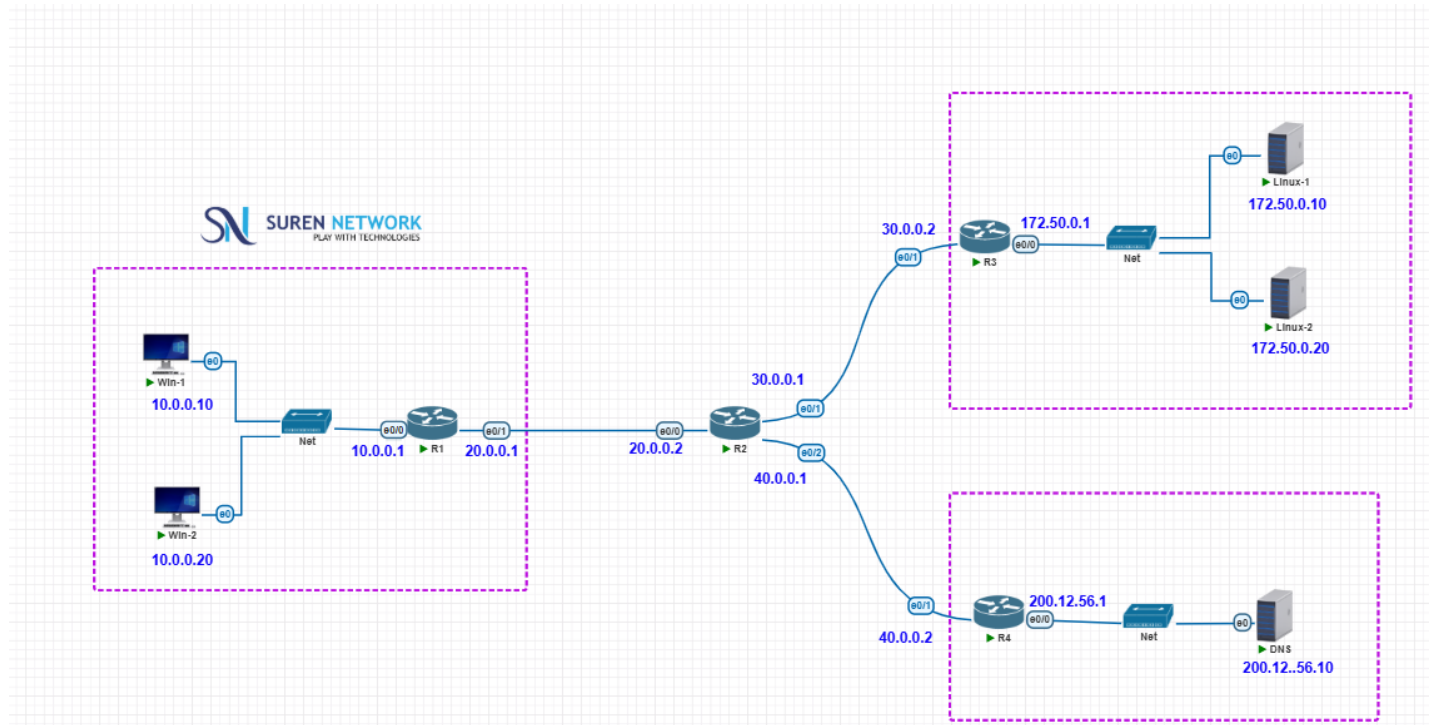


CCNA Lab: Access Control Lists & DNS

Standard ACL, Extended ACL, and DNS Name Resolution Walkthrough

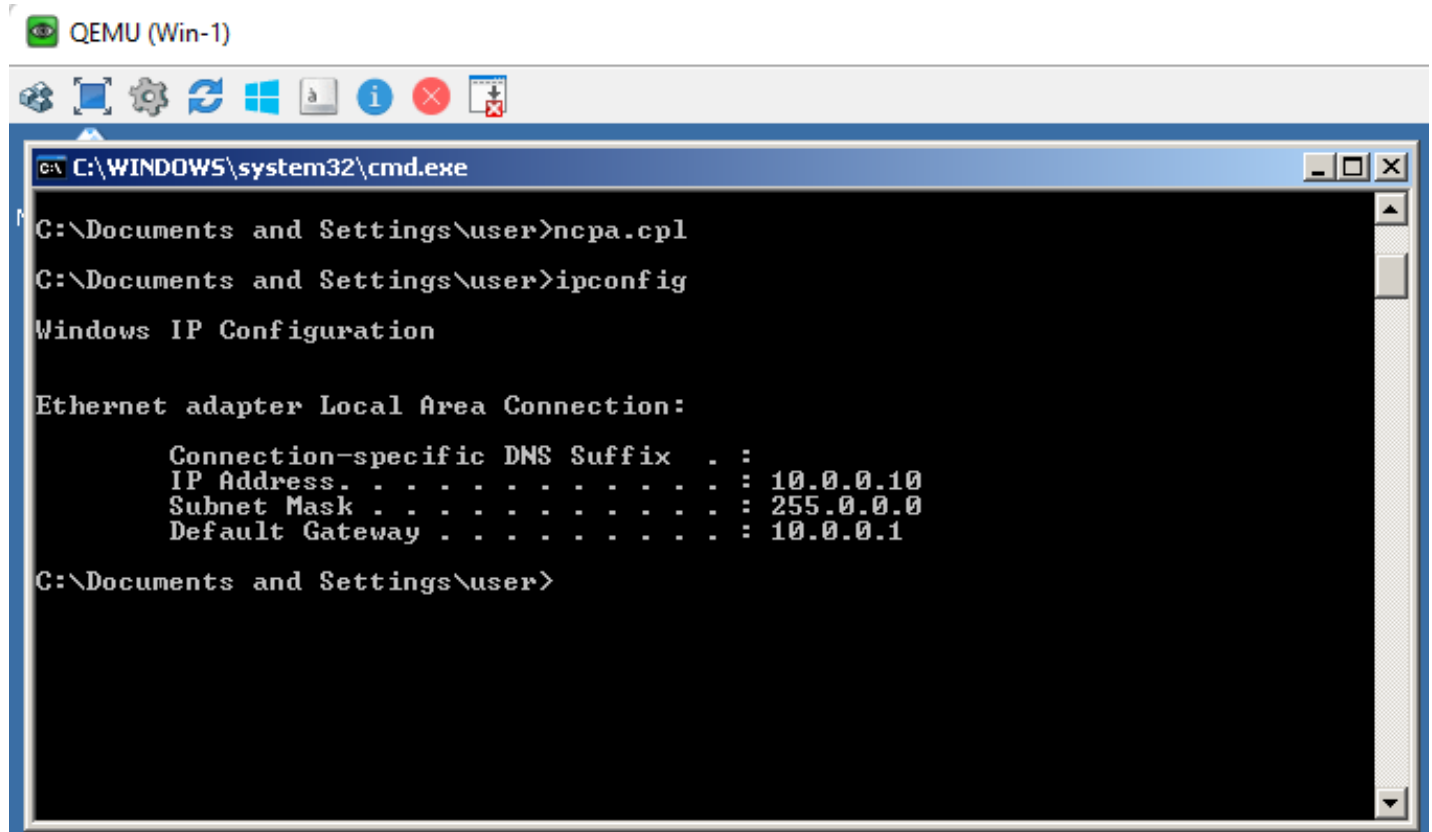
Compiled from lab screenshots with step-by-step explanations

1. Network Topology



Overall lab topology built in a Cisco simulator (Suren Network). R1 connects the LAN (Win-1 10.0.0.10, Win-2 10.0.0.20) via e0/0, and links to R2 over 20.0.0.0/8. R2 branches to R3 (172.50.0.0/8 network with Linux-1 and Linux-2 web servers) and R4 (200.12.56.0/8 network with a DNS server).

2. Win-1 IP Configuration



```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig
Windows IP Configuration

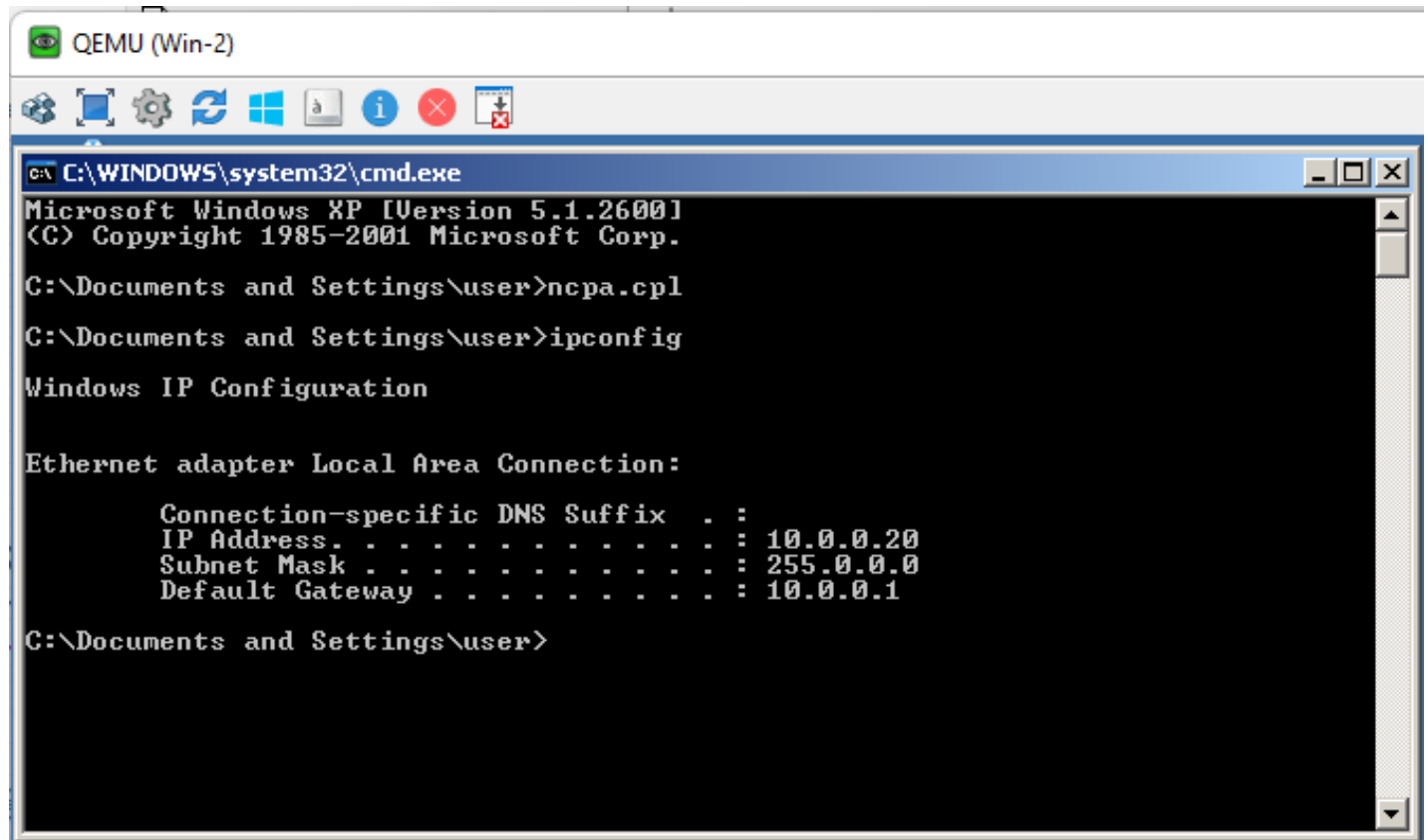
Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . .               : 10.0.0.10
    Subnet Mask . . . . .             : 255.0.0.0
    Default Gateway . . . . .         : 10.0.0.1

C:\Documents and Settings\user>
```

ipconfig output on Win-1 confirms IP address 10.0.0.10, subnet mask 255.0.0.0, and default gateway 10.0.0.1 (R1's e0/0 interface).

3. Win-2 IP Configuration



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

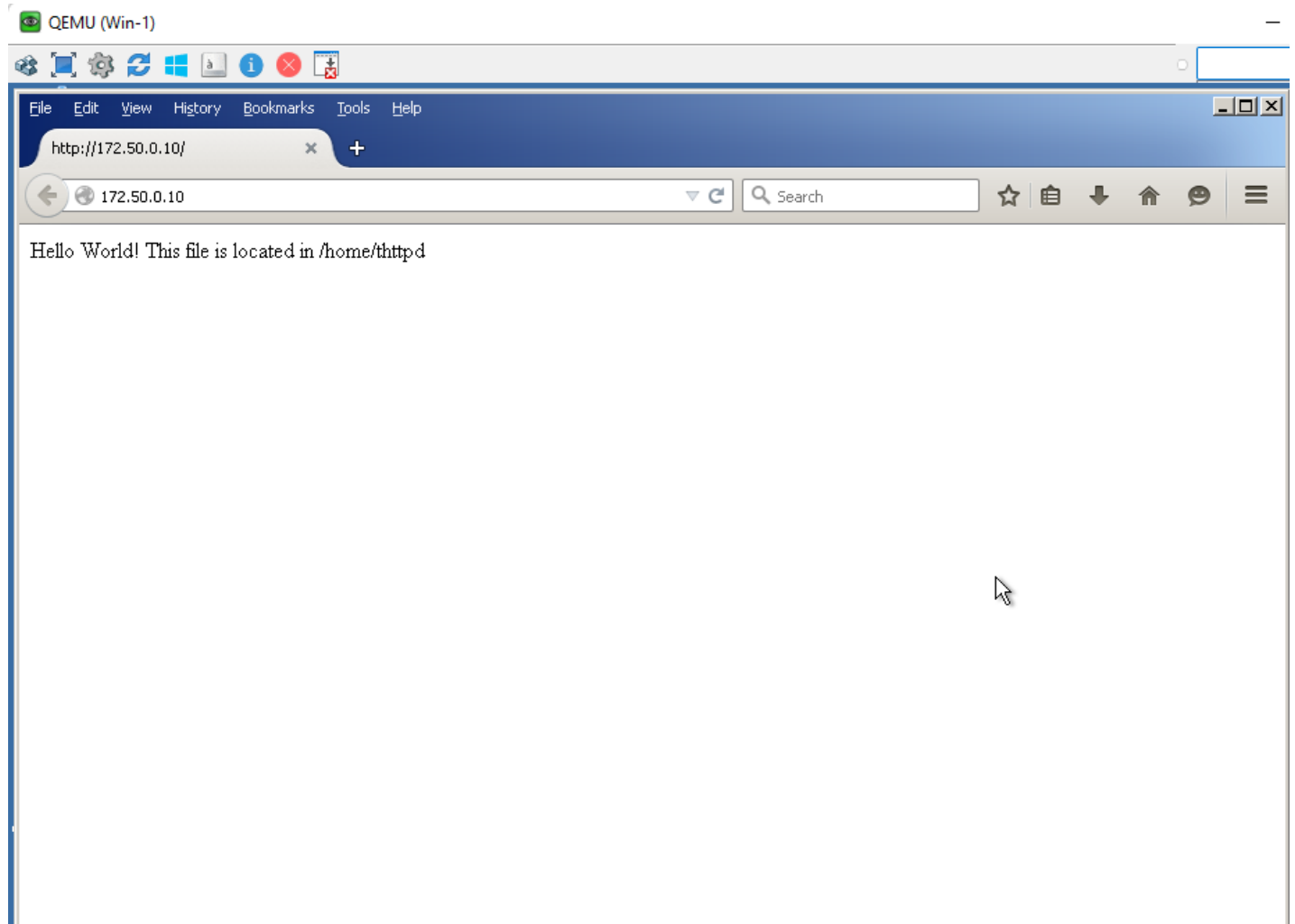
Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.20
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>
```

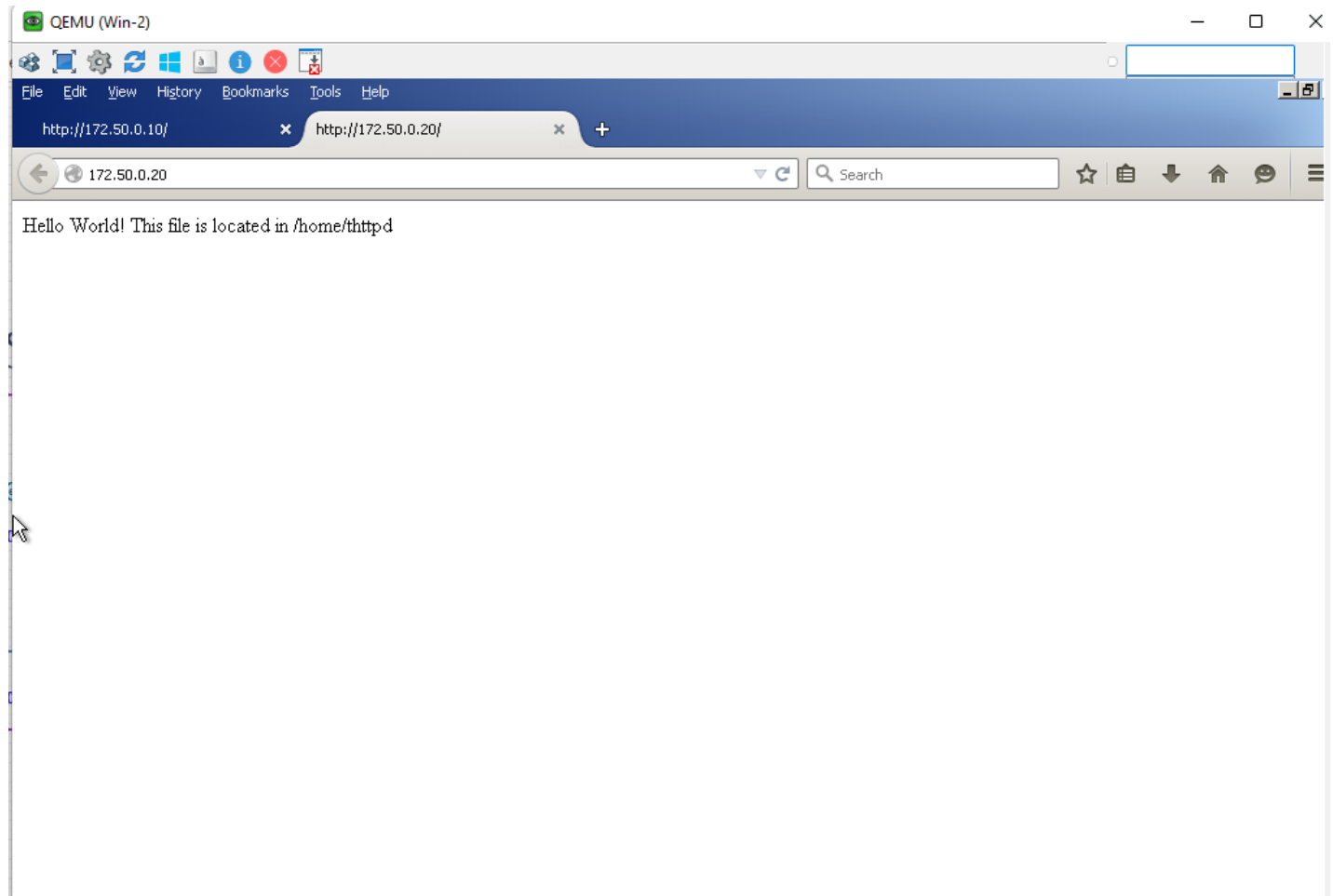
ipconfig output on Win-2 confirms IP address 10.0.0.20, subnet mask 255.0.0.0, and default gateway 10.0.0.1, verifying both LAN hosts are correctly addressed.

4. Win-1 Reaches Linux-1 Web Server



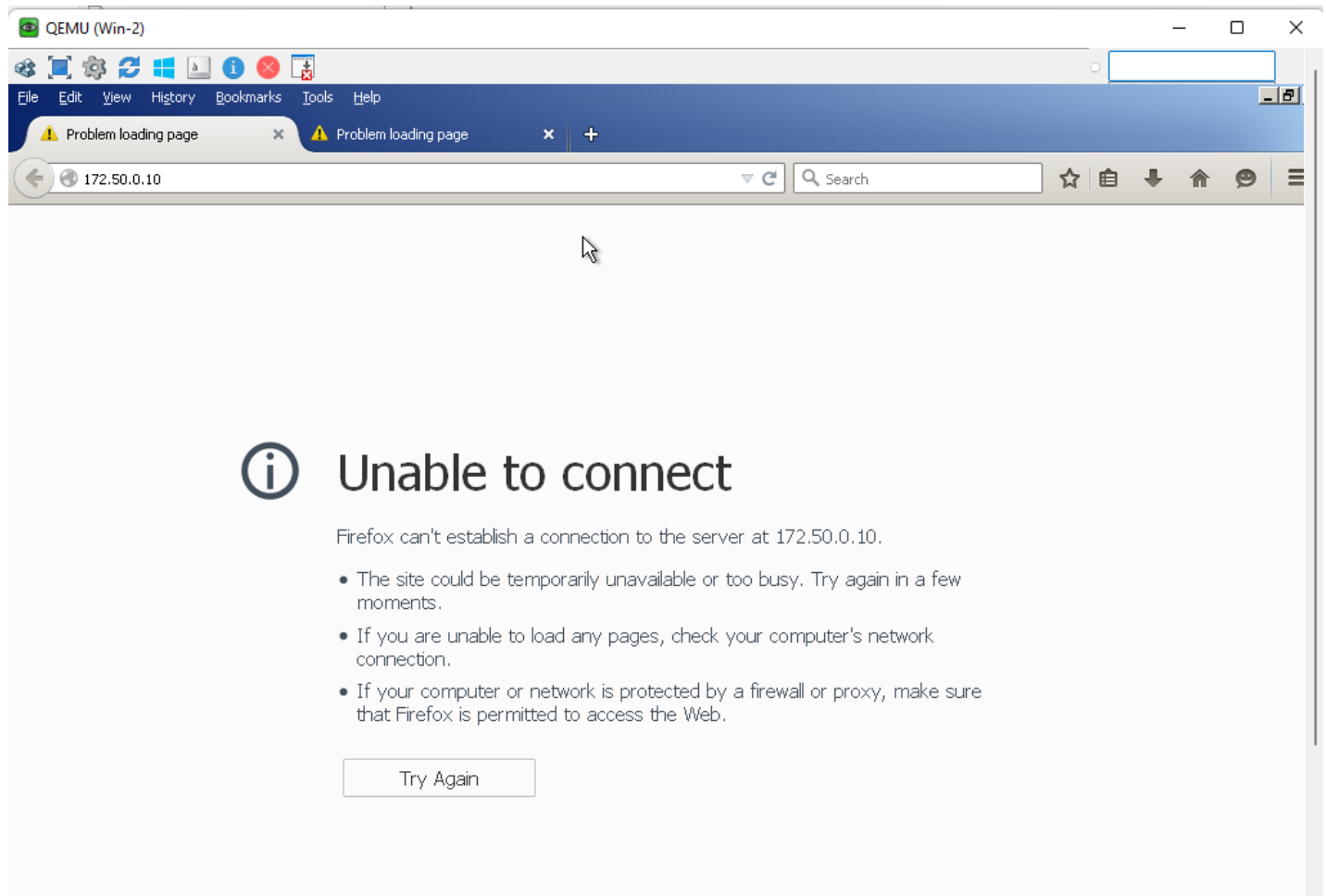
Before any ACL is applied, Win-1 successfully loads the 'Hello World!' page hosted on Linux-1 (172.50.0.10), confirming baseline end-to-end connectivity across R1 -> R2 -> R3.

5. Win-2 Reaches Both Web Servers



Win-2 opens both Linux-1 (172.50.0.10) and Linux-2 (172.50.0.20) in separate tabs, both returning the same 'Hello World!' test page, confirming full reachability before filtering is applied.

6. Win-2 Blocked From Linux-1 (After ACL 10 Applied to Win-1)



After the standard ACL is applied inbound on R1's e0/0 permitting only 10.0.0.10, Win-2's browser now fails to reach 172.50.0.10 with 'Unable to connect', showing the ACL is filtering traffic from 10.0.0.20.

7. Configuring Standard ACL 10 on R1 (First Attempt)

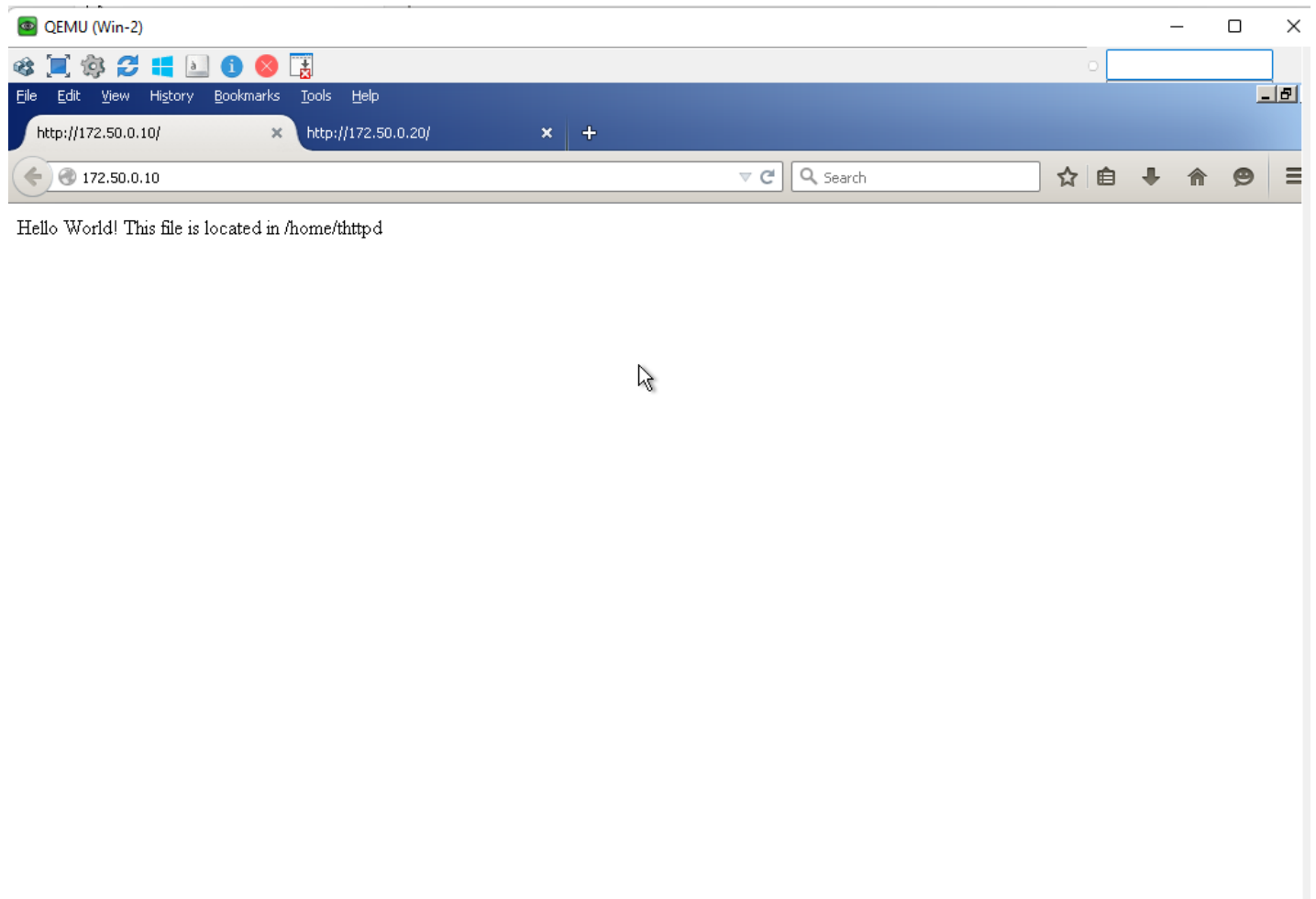
```
R1#
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 10 permit 10.0.0.10 0.0.0.0
R1(config)#int e0/0
R1(config-if)#ip access-group 10 in
R1(config-if)#
R1(config-if)#sh access-list
      ^
% Invalid input detected at '^' marker.

R1(config-if)#show access-list
      ^
% Invalid input detected at '^' marker.

R1(config-if)#exit
R1(config)#exit
R1#config t
*Sep 16 15:03:10.367: %SYS-5-CONFIG_I: Configured from console by console
R1#sh access-list
Standard IP access list 10
  10 permit 10.0.0.10
R1#
*Sep 16 15:04:17.703: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#
```

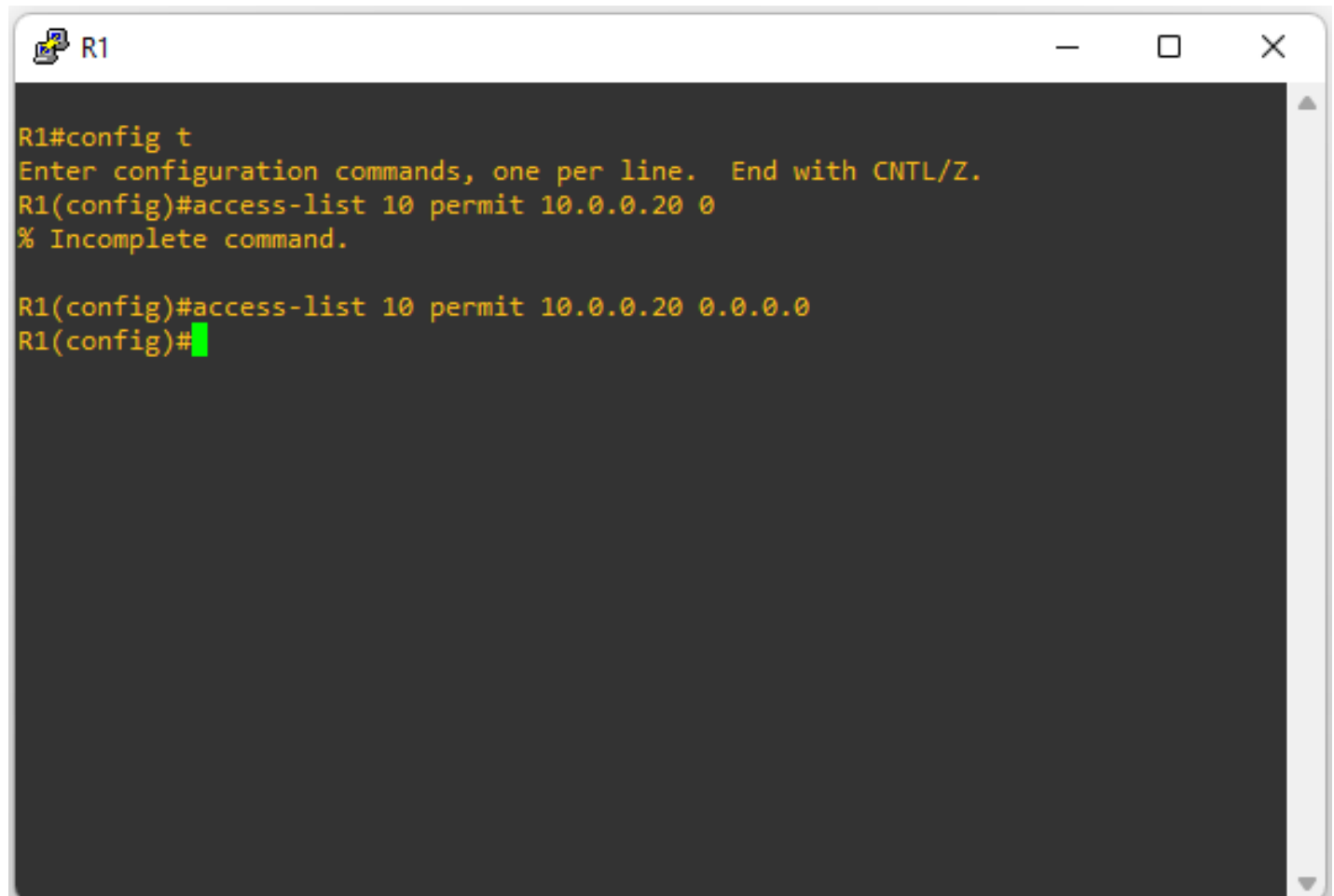
On R1: 'access-list 10 permit 10.0.0.10 0.0.0.0' is created and applied inbound on e0/0 with 'ip access-group 10 in'. The abbreviated 'sh access-list' command initially fails ('Invalid input'); the full 'show access-list' command also errors in that mode, but running 'show access-list' from privileged EXEC afterward correctly displays: 'Standard IP access list 10 - 10 permit 10.0.0.10'.

8. Win-1 Still Works After ACL 10



With ACL 10 (permit 10.0.0.10 only) applied inbound on e0/0, Win-1 (10.0.0.10) can still load <http://172.50.0.10/> successfully, since its source address matches the permit statement.

9. Adding a Second ACL Entry for Win-2

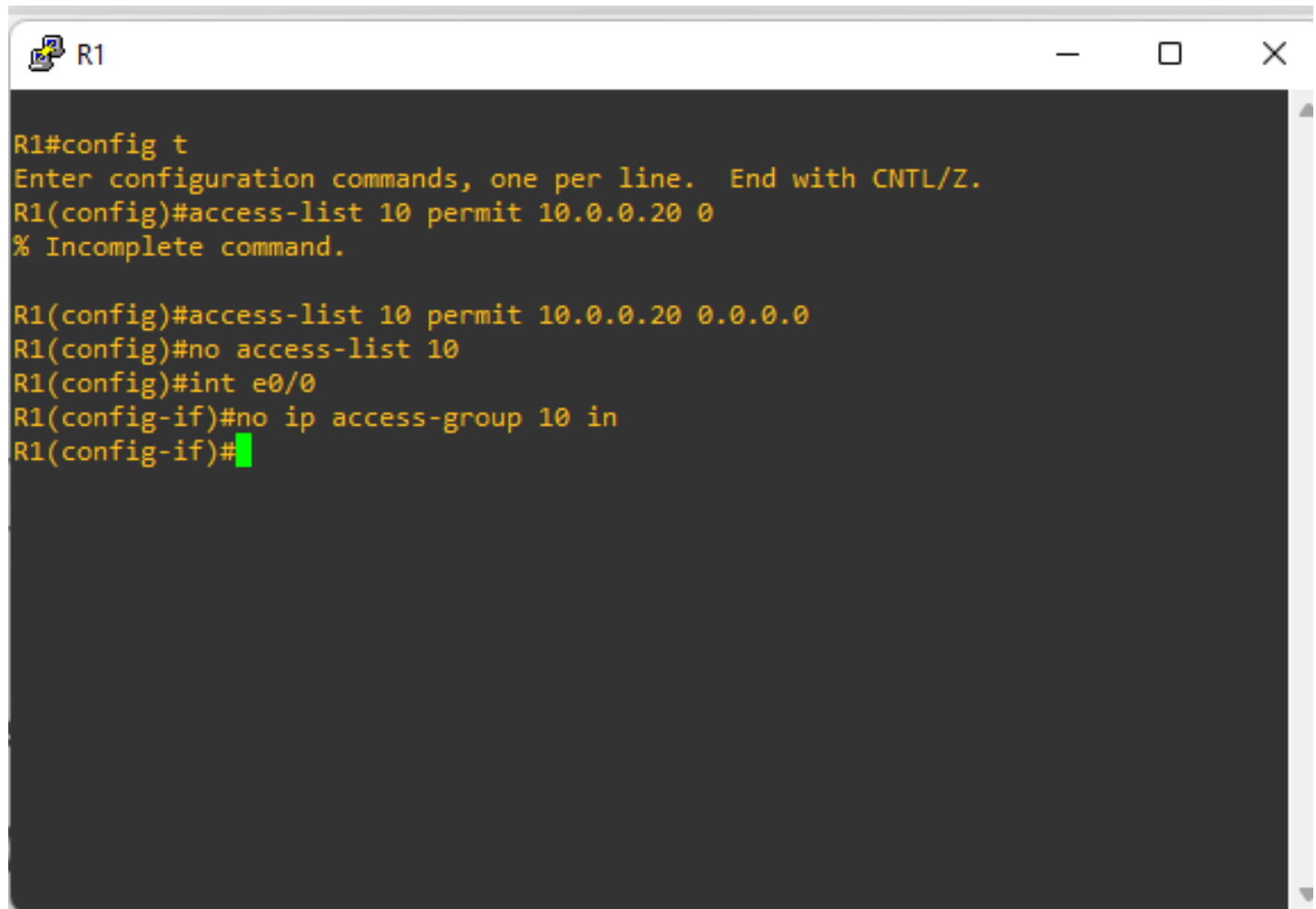


```
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 10 permit 10.0.0.20 0
% Incomplete command.

R1(config)#access-list 10 permit 10.0.0.20 0.0.0.0
R1(config)#
```

A second line is added: 'access-list 10 permit 10.0.0.20 0.0.0.0' (after an initial 'Incomplete command' error from omitting the wildcard mask), extending the standard ACL to also permit host 10.0.0.20.

10. Removing ACL 10 From the Interface

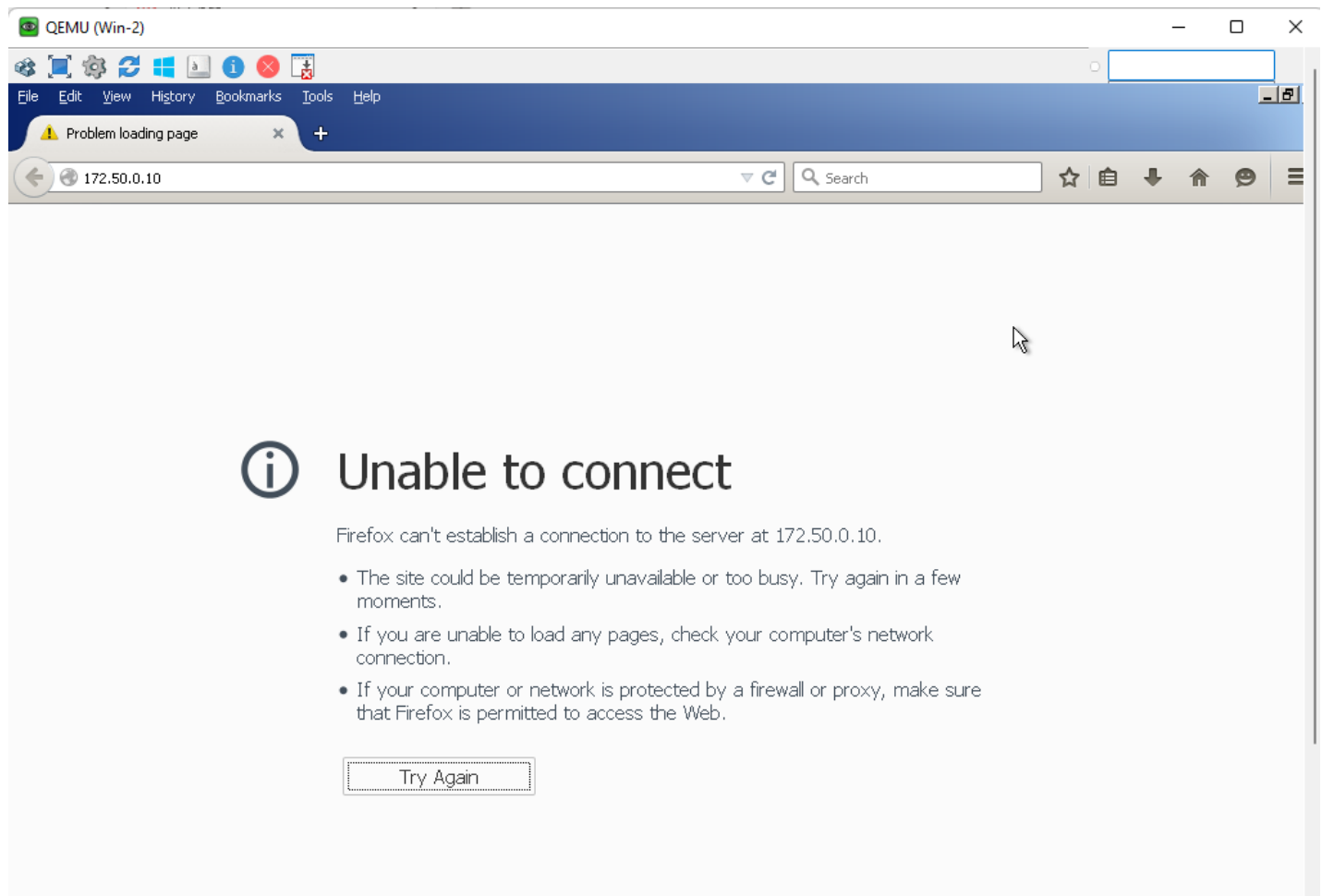


```
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 10 permit 10.0.0.20 0
% Incomplete command.

R1(config)#access-list 10 permit 10.0.0.20 0.0.0.0
R1(config)#no access-list 10
R1(config)#int e0/0
R1(config-if)#no ip access-group 10 in
R1(config-if)#
```

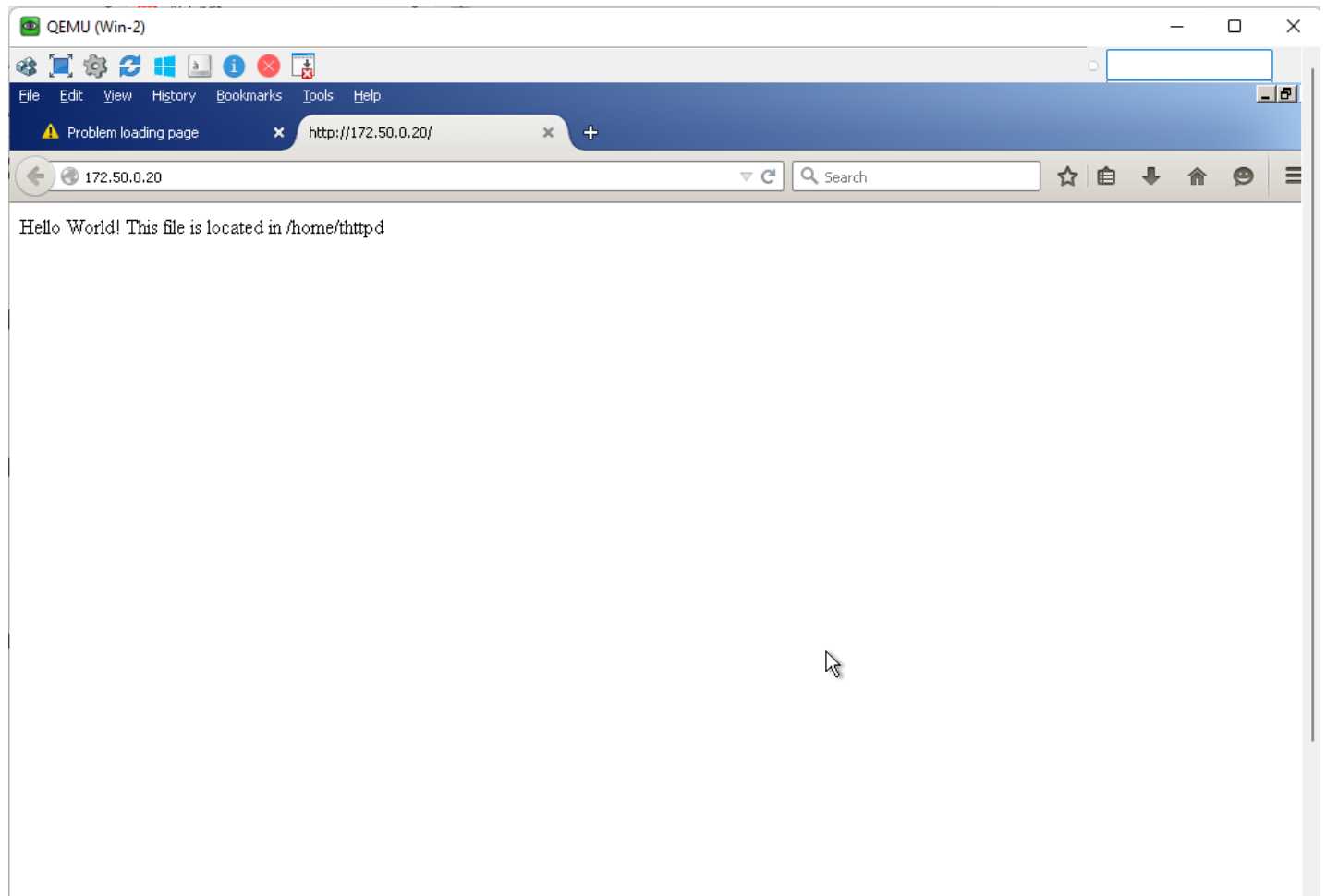
The access-list is removed with 'no access-list 10' and the ACL binding is removed from the interface using 'no ip access-group 10 in' on e0/0, restoring open access in preparation for the next test.

11. Win-2 Blocked Again (Retest)



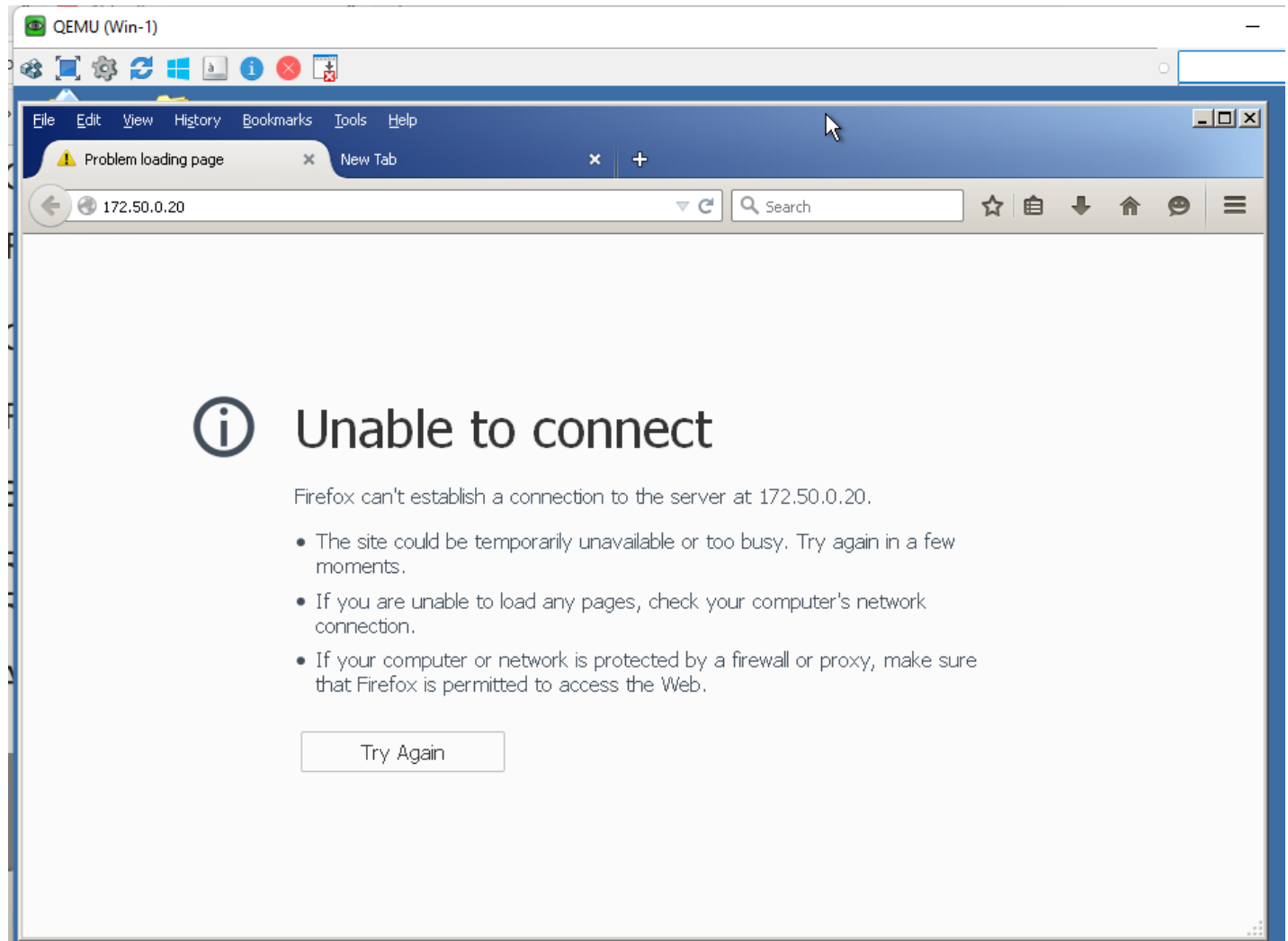
A further connectivity test shows Win-2 unable to reach 172.50.0.10, confirming filtering behavior at that stage of the exercise.

12. Win-2 Successfully Reaches Linux-2



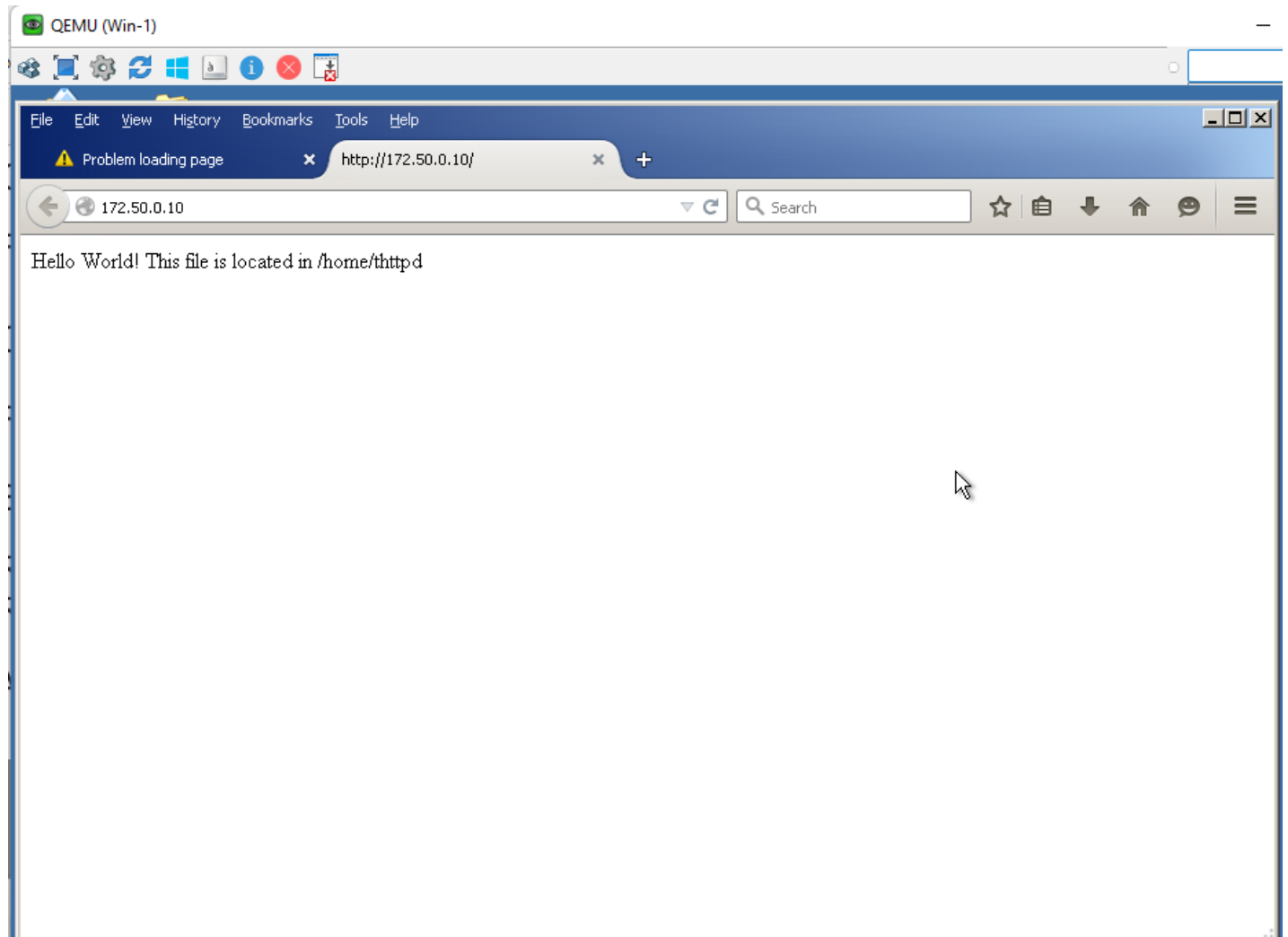
Win-2 successfully loads <http://172.50.0.20/> ('Hello World!'), showing that access to Linux-2 specifically remains permitted at this stage.

13. Win-1 Blocked From Linux-2



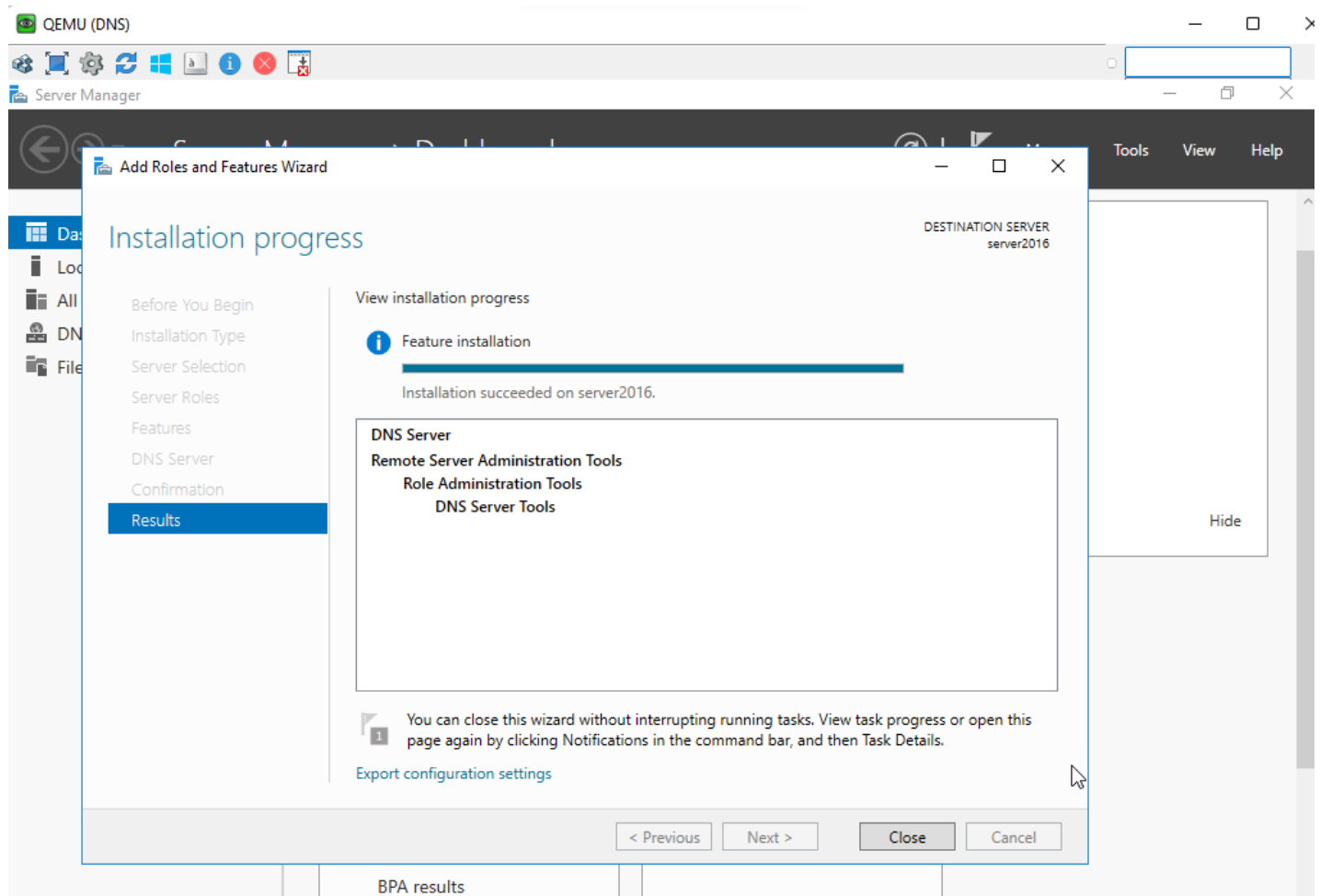
Win-1 attempts to reach 172.50.0.20 and gets 'Unable to connect', while its own tab for 172.50.0.10 still works — demonstrating host-specific filtering between the two Windows clients and the two Linux servers.

14. Win-1 Confirms Access to Linux-1



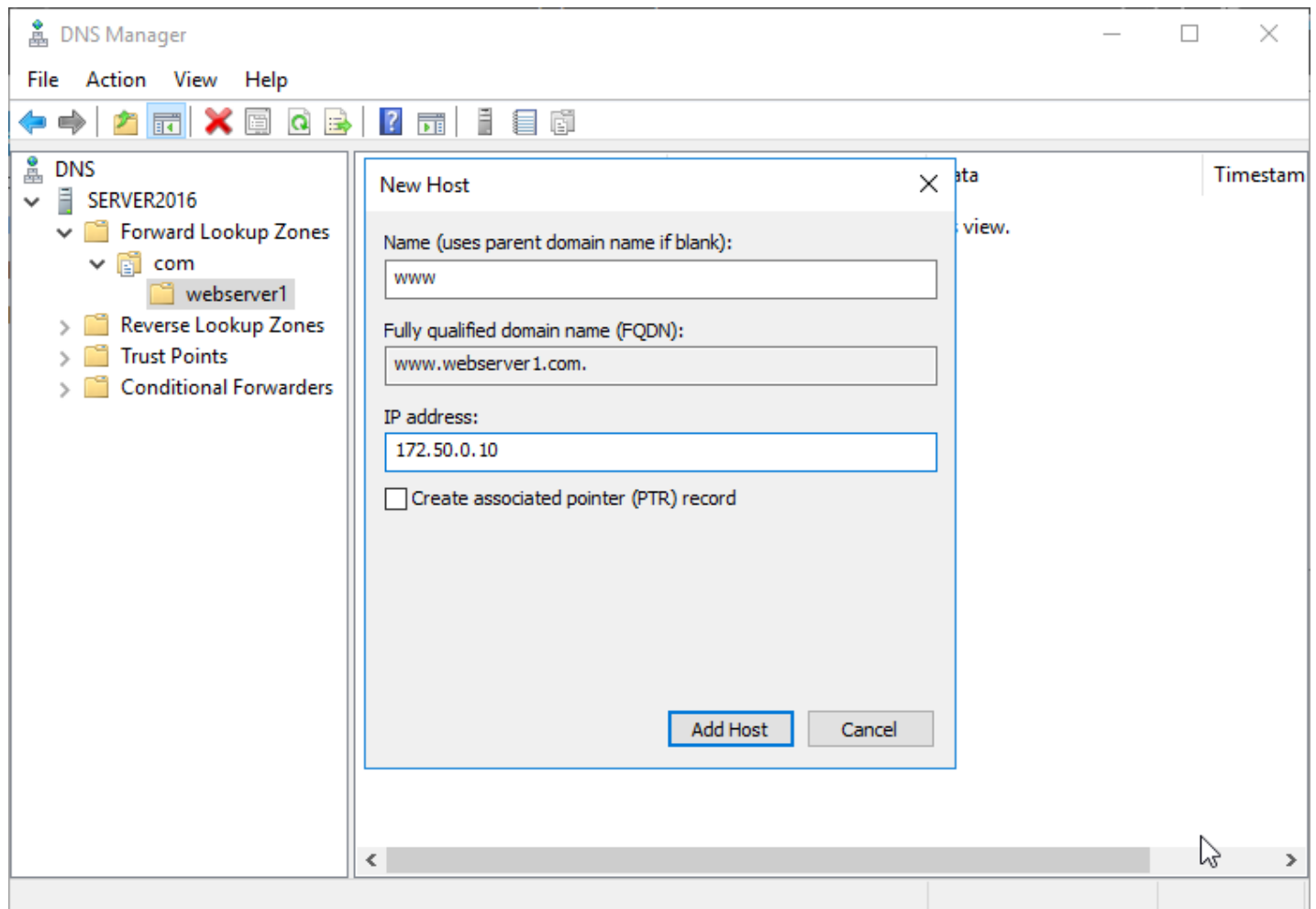
Win-1 reloads <http://172.50.0.10/> and receives the 'Hello World!' response, confirming its permitted path remains intact.

15. DNS Server Role Installation



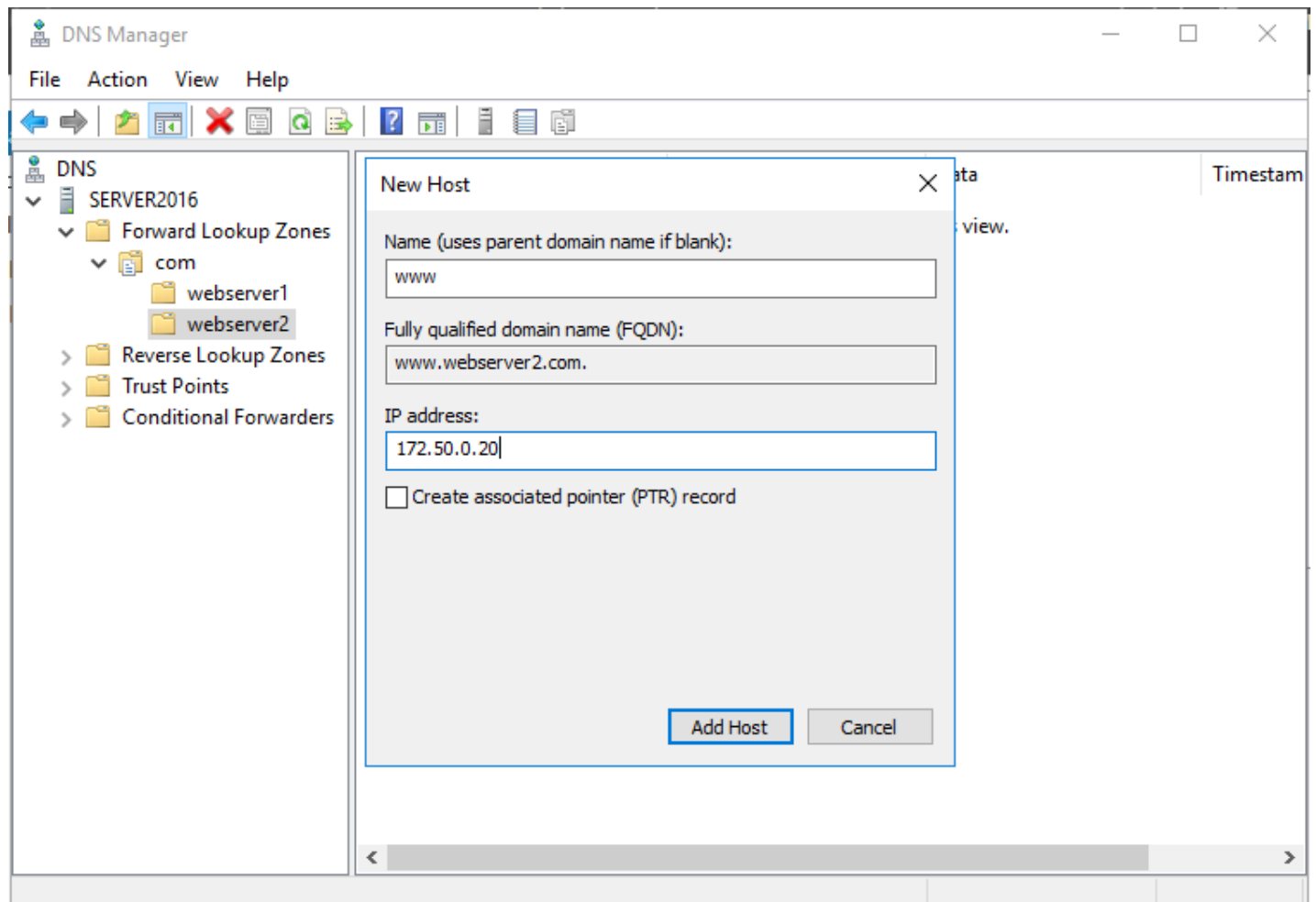
On the Windows Server 2016 DNS machine, the 'Add Roles and Features Wizard' completes installation of the DNS Server role along with Remote Server Administration Tools / Role Administration Tools / DNS Server Tools.

16. Creating DNS Host Record for webserver1.com



In DNS Manager, under Forward Lookup Zones -> com -> webserver1, a new host record 'www' is created resolving www.webserver1.com to 172.50.0.10 (Linux-1).

17. Creating DNS Host Record for webserver2.com



A second host record 'www' is created under the webserver2 zone, resolving www.webserver2.com to 172.50.0.20 (Linux-2).

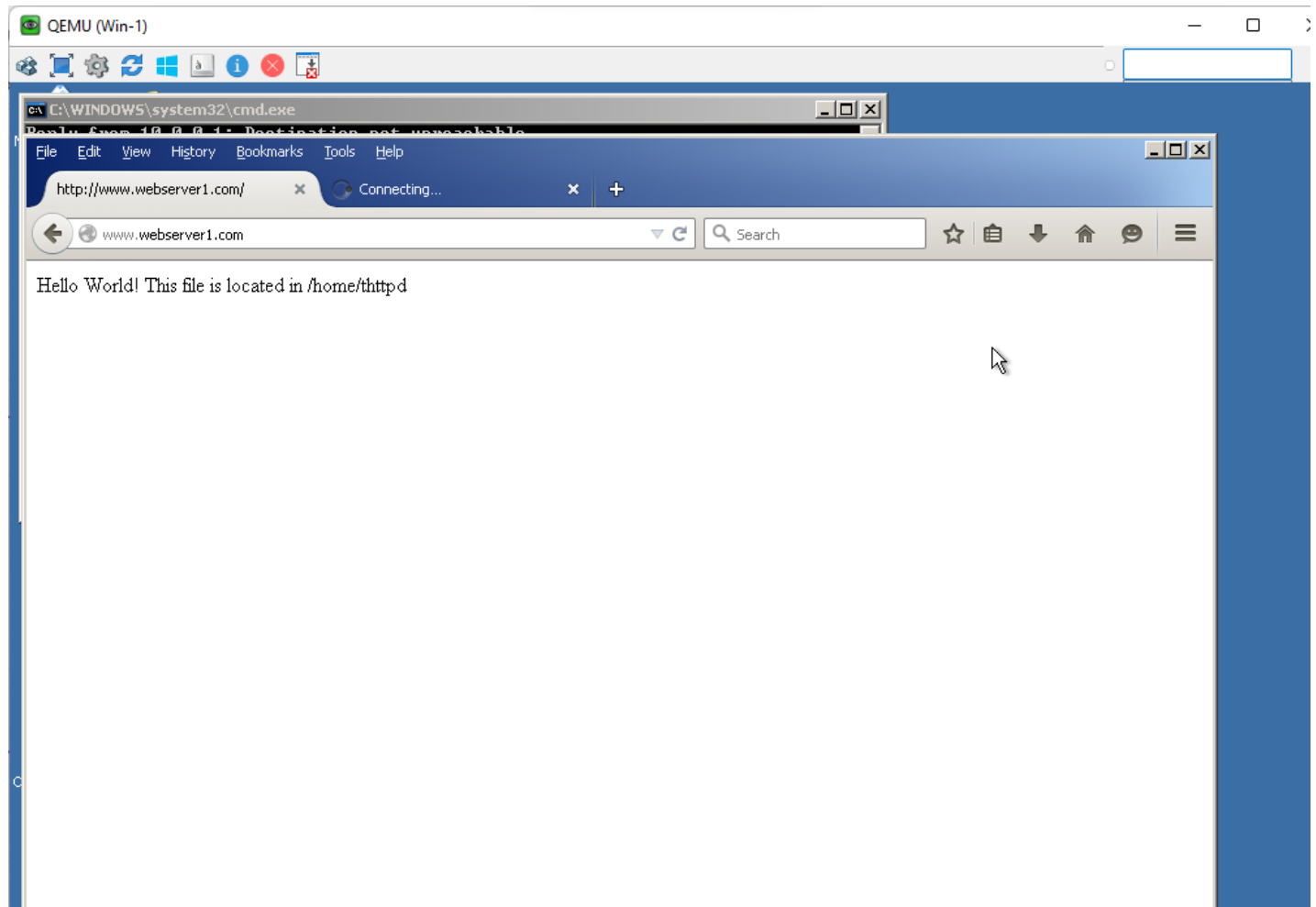
18. Configuring Extended ACL 100 on R1

 R1

```
R1(config-if)#
R1(config-if)#EXIT
R1(config)# 100 permit udp 10.0.0.0 0.255.255.255 200.12.56.10 0.0.0.0
R1(config)#exit
R1#
R1#
*Sep 16 16:48:14.461: %SYS-5-CONFIG_I: Configured from console by console
R1#sh access-list
Extended IP access list 100
 10 permit tcp host 10.0.0.10 host 172.50.0.10 eq www (45 matches)
 20 permit tcp host 10.0.0.20 host 172.50.0.20 eq www (30 matches)
 30 permit udp 10.0.0.0 0.255.255.255 host 200.12.56.10 (109 matches)
R1#
```

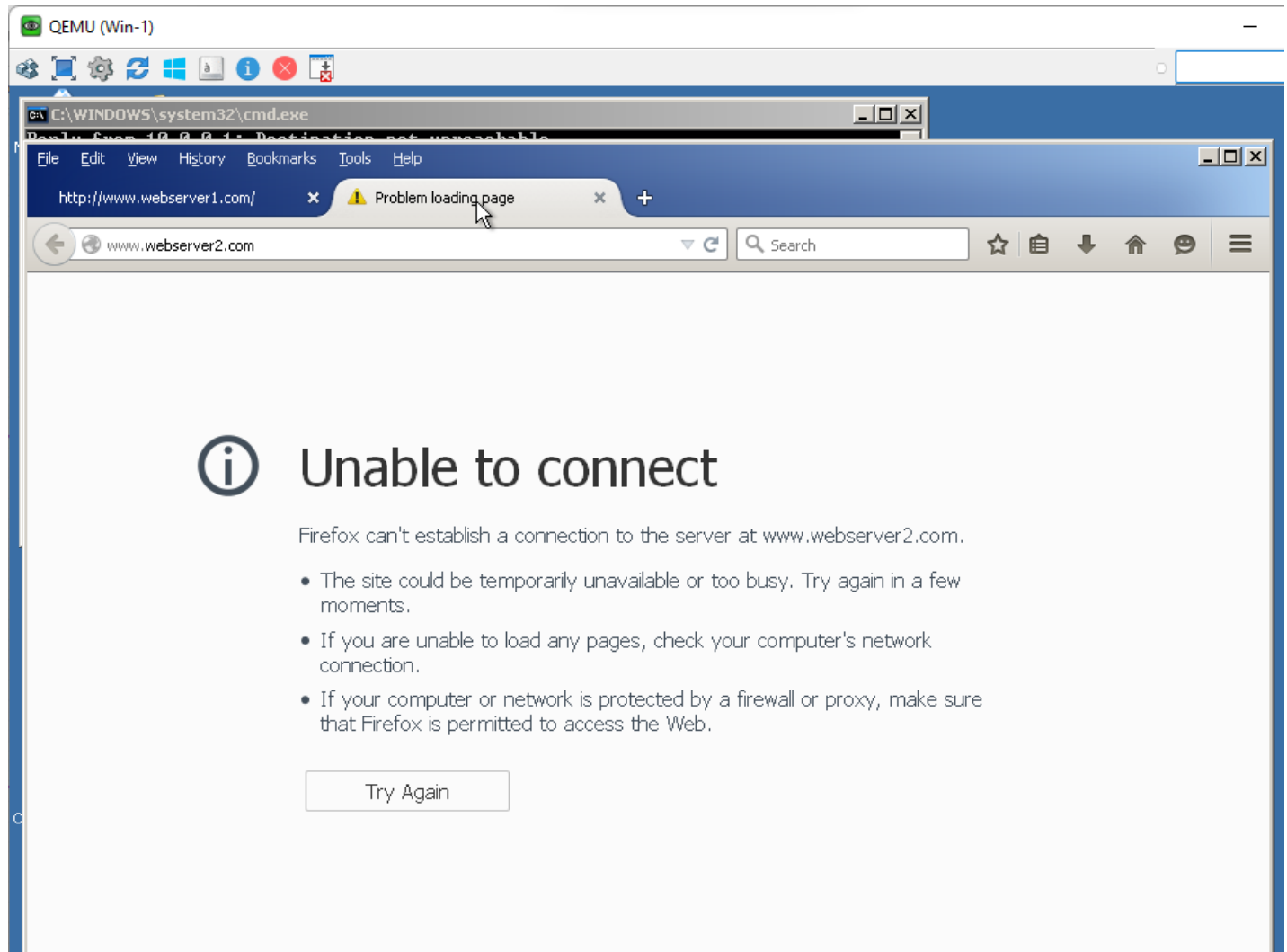
An extended ACL 100 is built on R1: permit tcp from host 10.0.0.10 to host 172.50.0.10 eq www; permit tcp from host 10.0.0.20 to host 172.50.0.20 eq www; and permit udp from the 10.0.0.0/8 network to host 200.12.56.10 (the DNS server), allowing DNS lookups. 'show access-list 100' confirms all three entries with hit counters (45, 30, and 109 matches respectively), showing the policy is actively being matched by traffic.

19. Win-1 Resolves and Loads www.webserver1.com



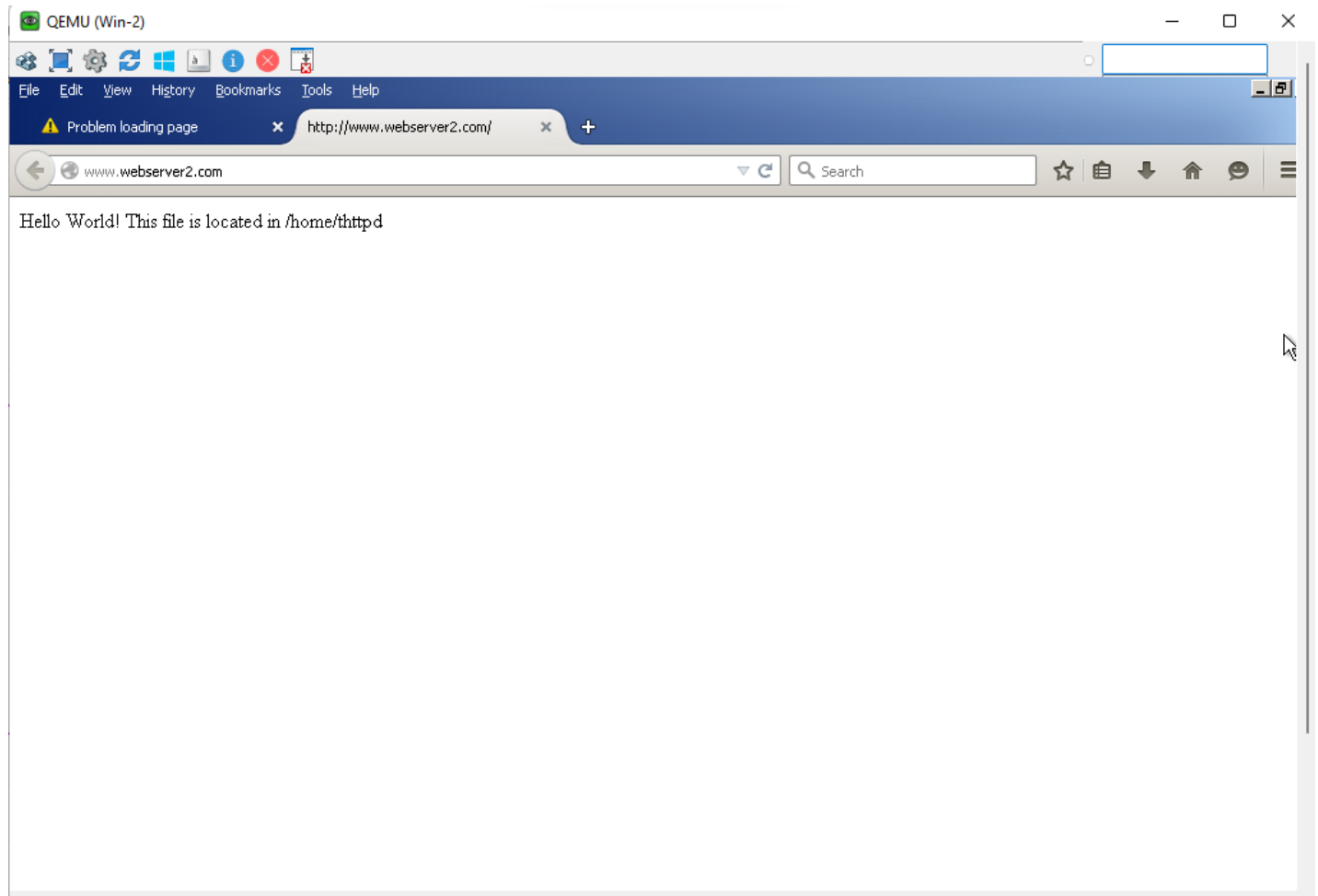
With DNS and the extended ACL in place, Win-1 browses to <http://www.webserver1.com/>, which resolves via DNS to 172.50.0.10 and loads the 'Hello World!' page — confirming DNS resolution plus the matching permit rule (10.0.0.10 -> 172.50.0.10 www) both work.

20. Win-1 Blocked From www.webserver2.com



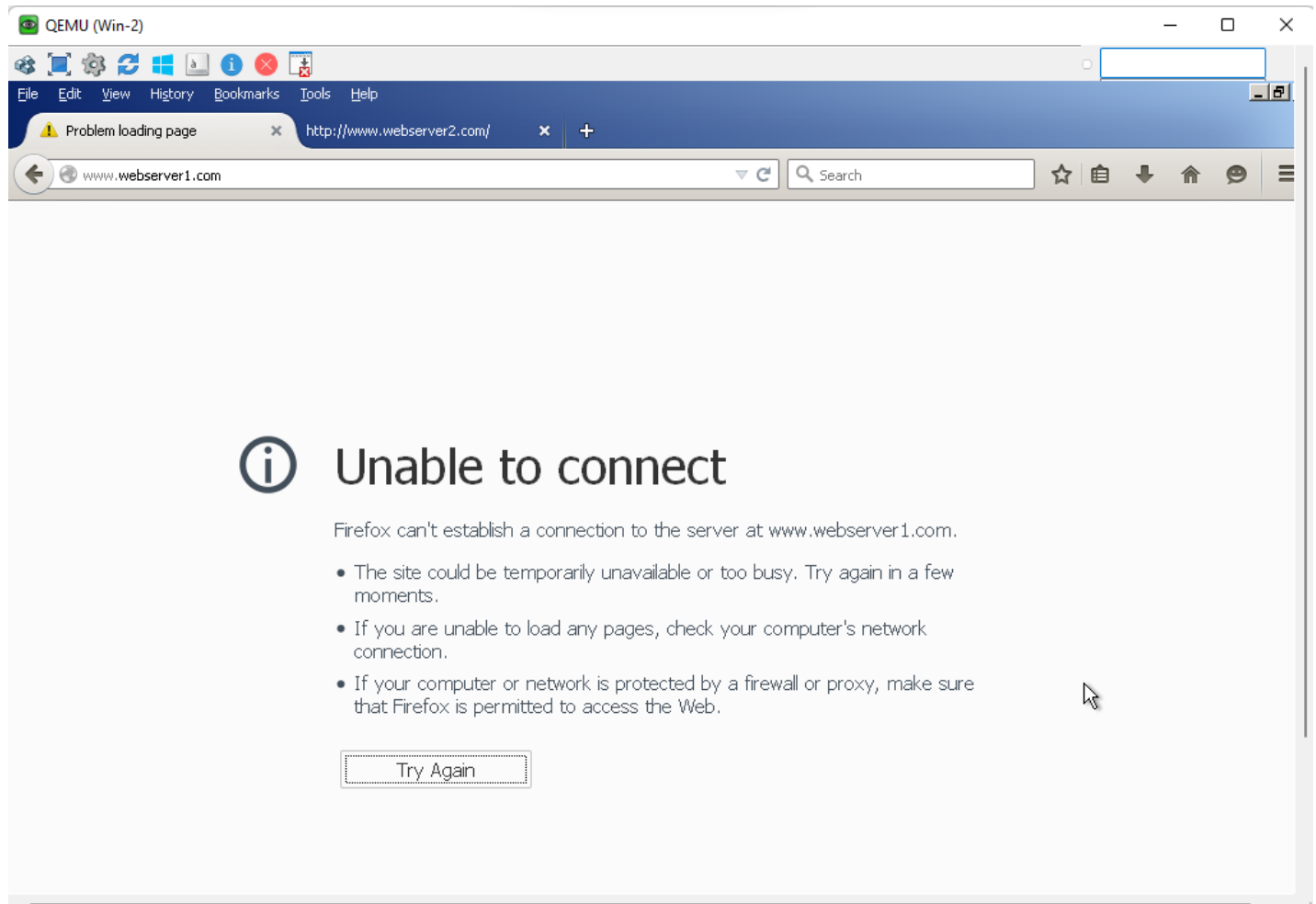
Win-1 then tries `http://www.webserver2.com/` (172.50.0.20) and fails with 'Unable to connect', since ACL 100 only permits 10.0.0.10 to reach 172.50.0.10, not 172.50.0.20 — demonstrating the extended ACL enforcing per-host, per-destination web access.

21. Win-2 Resolves and Loads www.webserver2.com



Win-2 successfully loads <http://www.webserver2.com/>, matching its permitted rule (10.0.0.20 -> 172.50.0.20 www).

22. Win-2 Blocked From www.webserver1.com



Win-2's attempt to reach <http://www.webserver1.com/> (172.50.0.10) fails with 'Unable to connect', confirming ACL 100 correctly restricts each Windows host to only its designated web server by domain name, completing the DNS + extended ACL verification.